

# Álvaro López Álvarez

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LinkedIn, GitHub, Portfolio

Full-Stack Developer with a Master's Degree in Computer Engineering (8.8/10) and two roles at the Instituto Tecnológico de Informática (ITI), spanning React/TypeScript front-end features, back-end integration, and LLM-driven workflow automation. Owns features end-to-end, from REST APIs and PostgreSQL schemas to deployed web apps, event-driven integrations, and containerized services with Docker. Keen to grow in cloud-native development on Microsoft Azure, having already learned Kafka, Spring Boot, and serverless patterns through self-directed projects. Works confidently in English (certified B2), Spanish, and Catalan.

## Core Competencies

- **Front-End Development**: TypeScript, JavaScript, HTML/CSS, React, Tailwind, Vite, Three.js
- **Back-End & APIs**: Golang (Gin), Java (Spring Boot), Python (Flask), REST APIs, JSON, OAuth 2.0/JWT
- **Databases & Integrations**: PostgreSQL, SQL, Firebase, Supabase, external service integration (Google Drive, Gemini, OpenAI APIs)
- **Cloud-Native & Event-Driven**: Docker, Docker Compose, serverless (FaaS), NATS JetStream, Apache Kafka, Apache APISIX, Vercel, Render
- **Engineering Fundamentals**: Git, GitHub Actions, debugging, testing (JUnit, Testcontainers, integration tests)

## Professional Experience

- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  *R&D Developer* — *Feb 2026–Jun 2026*
  - Implemented LLM-based improvements to a search engine
  - Built a LangGraph multi-step agent that automates a data-access workflow: it converts natural-language questions into validated, read-only SQL queries over a PostgreSQL data-governance schema to resolve data-access policies
  - Designed the workflow for safety and robustness: schema/table selection, structured SQL generation, execution validation, and retry-on-error
  - Developed the agent within an application for the management, governance, and sovereignty of industrial data

- **Instituto Tecnológico de Informática (ITI)** — **Valencia, España**
  *Full-Stack Developer Intern* — *Feb 2024–Jun 2024*
  - Developed and maintained the corporate personnel management web application (Intranet)
  - Implemented new front-end features using React and TypeScript, improving internal usability
  - Contributed to back-end development, ensuring smooth integration between services

## Selected Full-Stack & Cloud Projects

- **AIDucate** – *Go, Gin, PostgreSQL, React, TypeScript, Tailwind, Gemini API*: Full-stack SaaS platform generating personalized study plans, content, and exercises. JWT-authenticated Go/Gin REST API with PostgreSQL persistence and Gemini API integration; React/TypeScript frontend deployed on Vercel, backend on Render. *(In progress)*
- **FaaS with Golang** – *Go, Docker, NATS JetStream, Apache APISIX*: Distributed Function-as-a-Service (serverless) platform where Go workers execute arbitrary Docker-image functions. NATS JetStream work-queues for reliable, event-driven task distribution, Apache APISIX gateway for JWT auth and rate limiting, horizontal scaling via Docker Compose replicas.
- **Kafka + Spring Boot Ticketing Platform** – *Java, Spring Boot 3, Apache Kafka, Testcontainers*: Event-driven microservices (reservation, payment, notification, audit) using the outbox pattern, dead-letter topics, idempotent processing, and integration tests with Testcontainers.

- **CleanApp** – *React 19, TypeScript, Vite, Firebase*: Gamified household chore tracker for roommates with Firebase realtime database and authentication, algorithmic task rotation, and a points/leaderboard system.
- **OAuth RSE** – *Python, Flask, Docker, OAuth 2.0*: Dockerized task manager implementing the OAuth 2.0 Authorization Code flow with Google Drive API integration.

## Education

- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  *Master's Degree in Computer Engineering (MUIINF)* *2024–2026*
  Completed (Sep 2026), final grade 8.8/10. Topics: distributed systems, high-performance computing, cybersecurity, embedded systems.
- **Polytechnic University of Valencia (UPV)** **Valencia, España**
  *Grado (BSc) in Ingeniería Informática, Software Engineering track* *2020–2024*
  GPA: 8.88/10. Thesis: "ValenMove – Unified Public Transport App" (grade 9/10), a native Android app (Kotlin, MVVM) integrating public transport APIs, web scraping, and Open Data for real-time arrivals.

## Languages

- Spanish (native), Catalan (native), English (certified B2, confident conversational).

## References

- Available upon request.